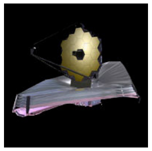

10799 - Interpreting JWST observations of high-z quasar environments with the BlueTides simulation

Cycle: 5, Proposal Category: AR

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OBSERVATIONS

ABSTRACT

The environments of high-redshift quasars are a key probe of the formation and growth of supermassive black holes (BHs) in the early Universe. JWST offers new opportunities to measure both their large- and local-scale environments. This gives an urgent challenge: how do we interpret these observations? From JWST, can we determine whether quasars lie in the most massive dark matter haloes? Are these quasars undergoing extreme growth due to galaxy-galaxy mergers?

In this project we will use the BlueTides simulation to develop JWST-specific theoretical predictions for the large- and local-scale environments of high-z quasars, by creating detailed mock NIRCam and NIRSpec imaging and spectroscopy. We will determine:

1. How observation biases affect the number of nearby galaxies observed on the Mpc scale
2. How well observable properties (galaxy number counts, infall velocities) can determine halo mass
3. Whether BlueTides reproduces the quasar large-scale environments seen with JWST, testing the BH growth model
4. The fraction of BlueTides quasars merging with nearby (kpc-scale) companion galaxies, and how this compares to JWST-observed quasars
5. How these local quasar environments link to the BH growth.

These results will be directly applicable to a large number of NIRCam and NIRSpec IFU observations studying large- and local-scale quasar environments, as well as more general studies of galaxy overdensities and merger rates. Our work will provide a framework for understanding these JWST observations in the broader context of BH growth, and test the current models of galaxy and BH formation physics, which will lead us closer to understanding galaxy evolution more broadly.

OBSERVING DESCRIPTION

This is a theory proposal to interpret the large- and local-scale environments of high-z quasars observed with JWST.